

Interface for Observed Passenger Flow Data for Schedule-Based Assignment Models

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1 Purpose

This document specifies a **standard interface** for providing observed passenger flow data to be used in the estimation and validation of a schedule-based public transport assignment model.

The objective is to allow analysts to provide observed flows in a way that is:

- consistent with the **scenario representation** (stops, trips, timetable),
- independent of **internal assignment indexing conventions**,
- flexible enough to support different types of observations,
- robust to **missing data** and heterogeneous measurement systems,
- easily extendable to likelihood-based estimation.

This interface describes only the *data format*. The mapping between observations and model outputs, as well as the likelihood specification, will be handled by separate components.

2 Design principles

2.1 Separation from assignment internals

The assignment algorithm operates on time-expanded graphs and internal link indices. These indices must remain *internal* to the model implementation. Analysts providing data should not need to know them.

Therefore, observations are expressed exclusively in terms of:

- scenario-level identifiers (stop, line, trip),
- timetable times,
- measurement metadata.

2.2 Event-based timing

Time bins in the scenario represent desired departure times and are not tied to timetable events. Observations of passenger flows, however, are associated with **events in the timetable** (arrivals, departures, vehicle loads).

Hence:

- All observed flows are linked to timetable events.
- Time is expressed as clock time (HH:MM:SS).
- Observations refer to time windows on the service day.

2.3 Partial and heterogeneous data

Not every link in the network has data. Observations may include:

- vehicle load counts,
- boarding counts,
- alighting counts,
- aggregated measurements.

The interface must therefore:

- allow missing data naturally,
- support multiple measurement methods,
- remain simple for analysts to fill in.

2.4 Separation of data and statistical assumptions

The data file should contain *observations only*. It should not include statistical parameters such as variances or weights.

Instead, each observation is tagged with a `method_id`. The likelihood wrapper will interpret this identifier and assign the appropriate measurement model.

3 File format

Observed flows are provided in a table file:

- `observations.csv` (recommended), or
- Parquet with identical schema.

Each row represents one measurement.

4 Columns

4.1 Identification and metadata

Column	Type	Description
<code>obs_id</code>	string	Unique identifier of the observation.
<code>method_id</code>	string	Identifier of measurement method or sensor.
<code>obs_type</code>	string	Type of observation. Currently: <code>instant_load</code> .
<code>value</code>	float	Observed passenger count. Must be ≥ 0 .

4.2 Time window

Column	Type	Description
<code>time_start</code>	HH:MM:SS	Start of observation window.
<code>time_end</code>	HH:MM:SS	End of observation window (exclusive).

Interpretation: the measurement refers to timetable events within `[time_start, time_end)`. If an observation is instantaneous, a short window (e.g. 30–60s) should be used.

4.3 Location and event

Each observation must identify where in the timetable it applies.

Trip-based mode (preferred)

Column	Type	Description
<code>trip_id</code>	string	Trip identifier from the timetable.
<code>stop_id</code>	string	Stop identifier.
<code>event</code>	string	ARR or DEP.

Line-based mode (aggregate)

Used when trip identity is unknown.

Column	Type	Description
<code>line_id</code>	string	Line identifier.
<code>stop_id</code>	string	Stop identifier.
<code>event</code>	string	ARR or DEP.

Rule: exactly one of `trip_id` or `line_id` must be provided.

4.4 Optional metadata

Column	Type	Description
<code>campaign_id</code>	string	Measurement campaign.
<code>source</code>	string	Sensor or file source.
<code>notes</code>	string	Free-text notes.
<code>stop_sequence</code>	int	Optional disambiguation if a trip visits a stop multiple times.

5 Instantaneous load semantics

For `obs_type = instant_load`, the value represents observed onboard passengers near a timetable event.

The exact interpretation (e.g. load after departure vs. before arrival) is defined by the measurement method and handled by the mapping layer.

6 Validation rules

The loader should verify:

- `obs_id` unique.
- `value` ≥ 0 .
- Valid time format HH:MM:SS.
- `time_end` $>$ `time_start`.

- `stop_id` exists in scenario.
- `trip_id` or `line_id` exists.
- Exactly one of them is provided.
- `event` in `{ARR, DEP}`.

7 Example

```
obs_id,method_id,obs_type,value,time_start,time_end,trip_id,line_id,stop_id,event
L001,APC_V1,instant_load,37,08:00:00,08:02:00,T12,,A,DEP
L002,APC_V1,instant_load,22,08:10:00,08:12:00,T12,,B,ARR
L101,MANUAL_V1,instant_load,18,17:30:00,17:45:00,,L1,C,DEP
```

8 Future extensions

The interface is designed to support:

- boarding counts,
- alighting counts,
- link flows,
- likelihood-based estimation.

Mapping from observations to model quantities will be handled by dedicated mapper components, using:

- the scenario,
- the assignment outputs,
- the `method_id`.